

Probing Physical-Parameter Identifiability from Motion

Observation $\times$ Excitation, via Differentiable MPM + 3D Gaussian Splatting

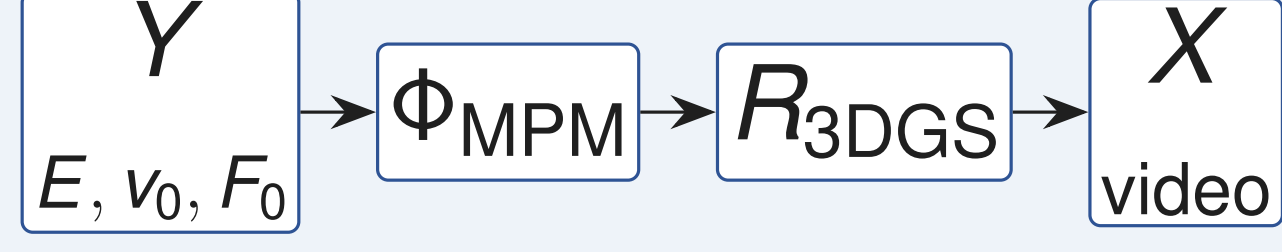
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The question & three regimes

When is a physical parameter recoverable from how an object moves?

We probe this along **two orthogonal axes** over the differentiable map

$g = R_{3DGS} \circ \Phi_{MPM}$:

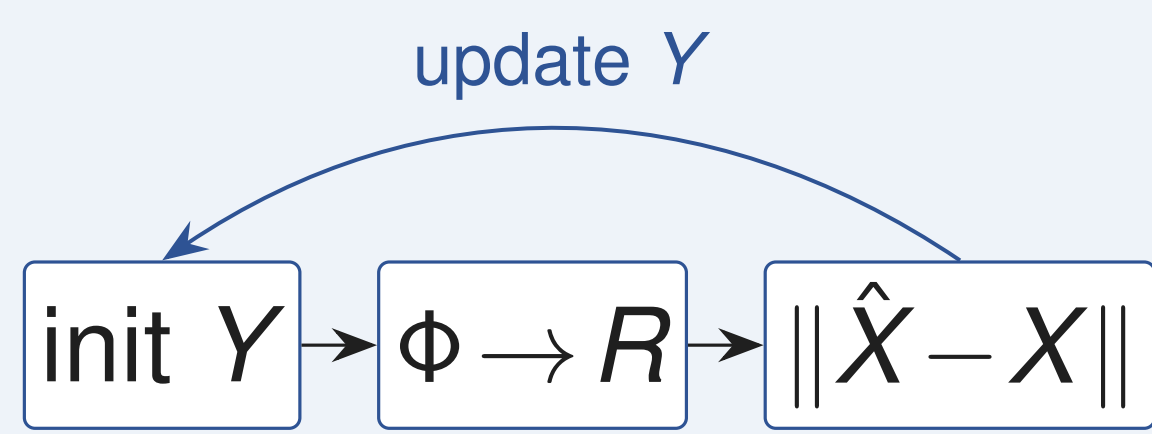


Three supervision regimes, kept honestly separate:

- **image loss** on pixels — **recoverable from video** (Pillar A).
- **trajectory loss** — needs GT motion, *oracle* only.
- **forward + finite difference** — synthetic identifiability analysis (Pillar B).

Pillar A — Observation geometry (image loss, recoverable from video)

Recovery loop. Freeze the 3DGS scene, leave only Y free, backprop the image loss through render + a differentiable MPM solver.

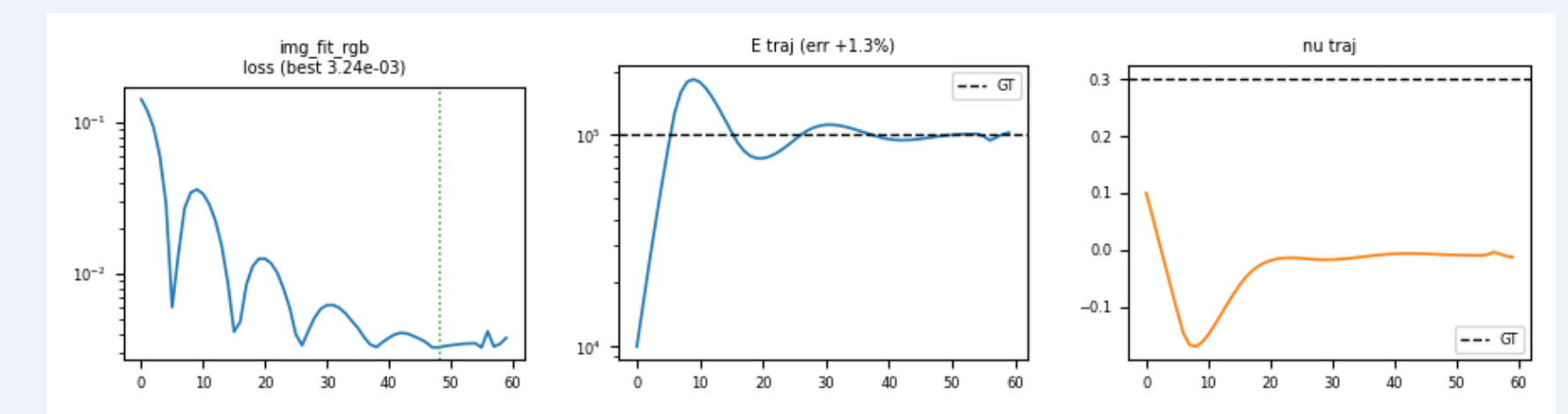


Two levers: *observation* (#views) and *excitation design*. Y scales from a global scalar E to per-particle fields.

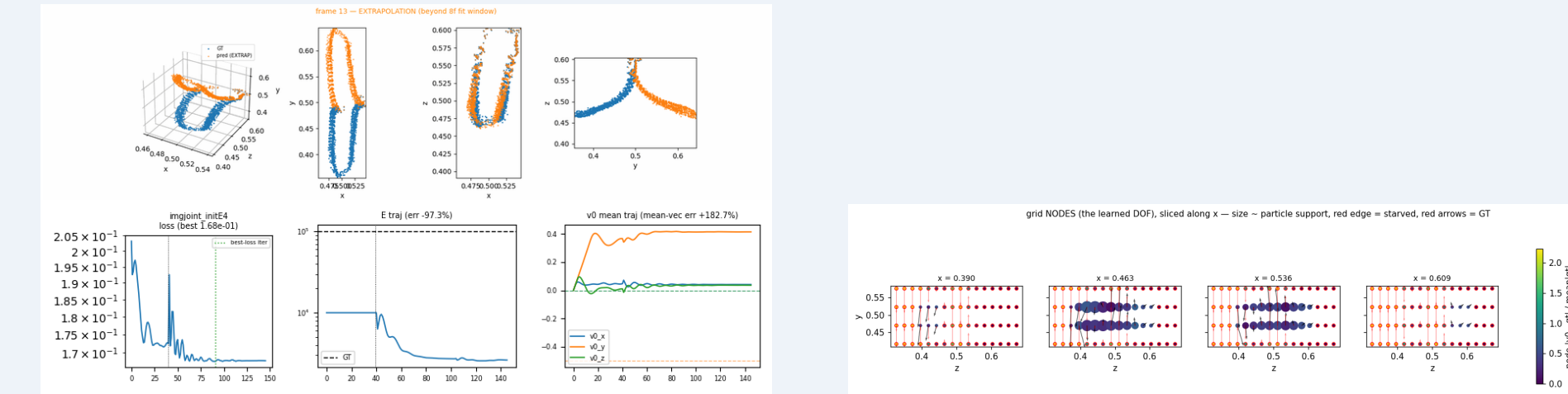
A second view breaks the $E-v_0$ degeneracy

	known v_0	unknown (E, v_0)
single view	E 1.31% ✓	$E \rightarrow 2742$ (−97%) ✗
multi view	—	E 1.04%, v_0 1.12% ✓

One view cannot separate “stiffer + faster” from “softer + slower” (they render alike), so the joint fit **collapses**; a side view supplies the missing constraint (~ 13 min). **Excitation design** likewise gates a recoverable v_0 field: a *bend* gives rel- L_2 0.26 vs a harder *mid-kick* 0.84.



A. known $v_0 \Rightarrow E$ 1.31%



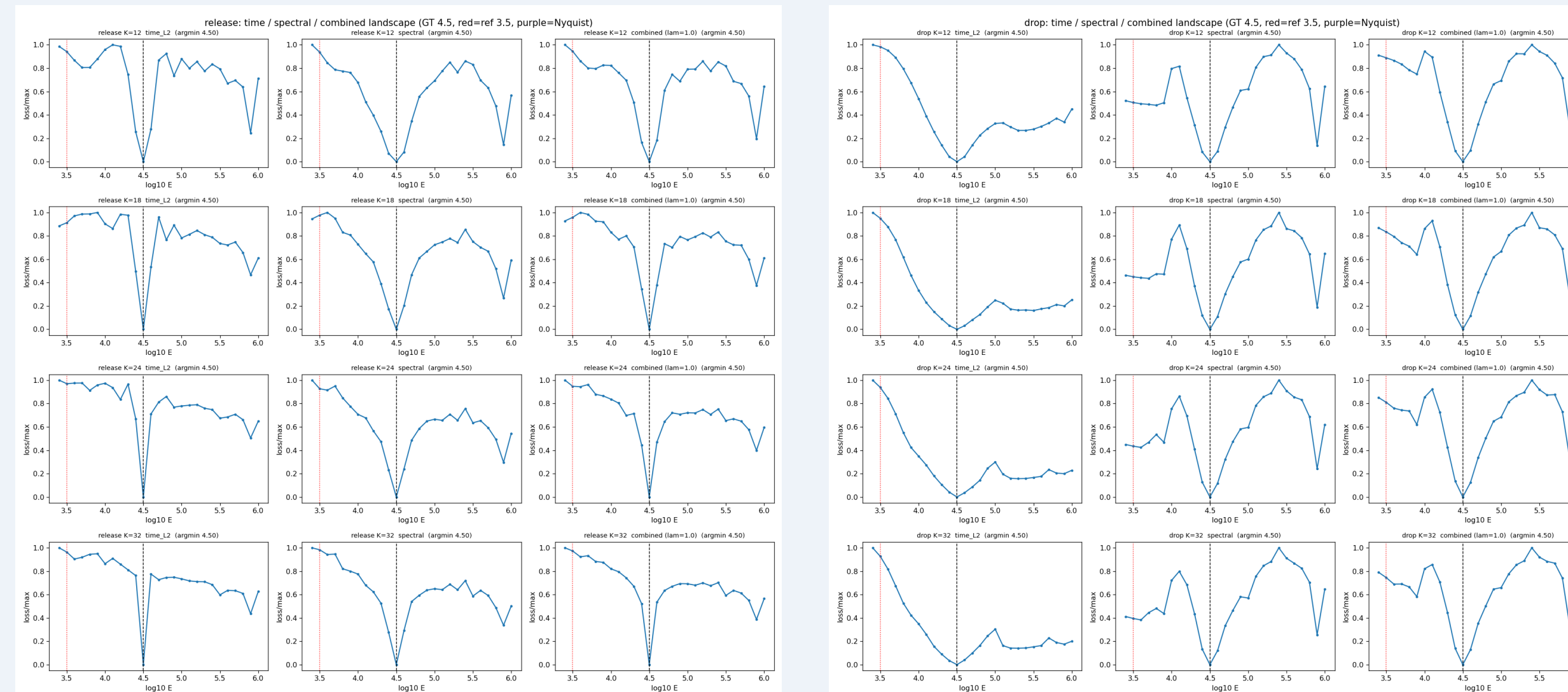
B./C. single fail vs multi solve · D. v_0 bend field

Pillar B — Excitation design & the two E channels (forward + finite-difference analysis; synthetic —

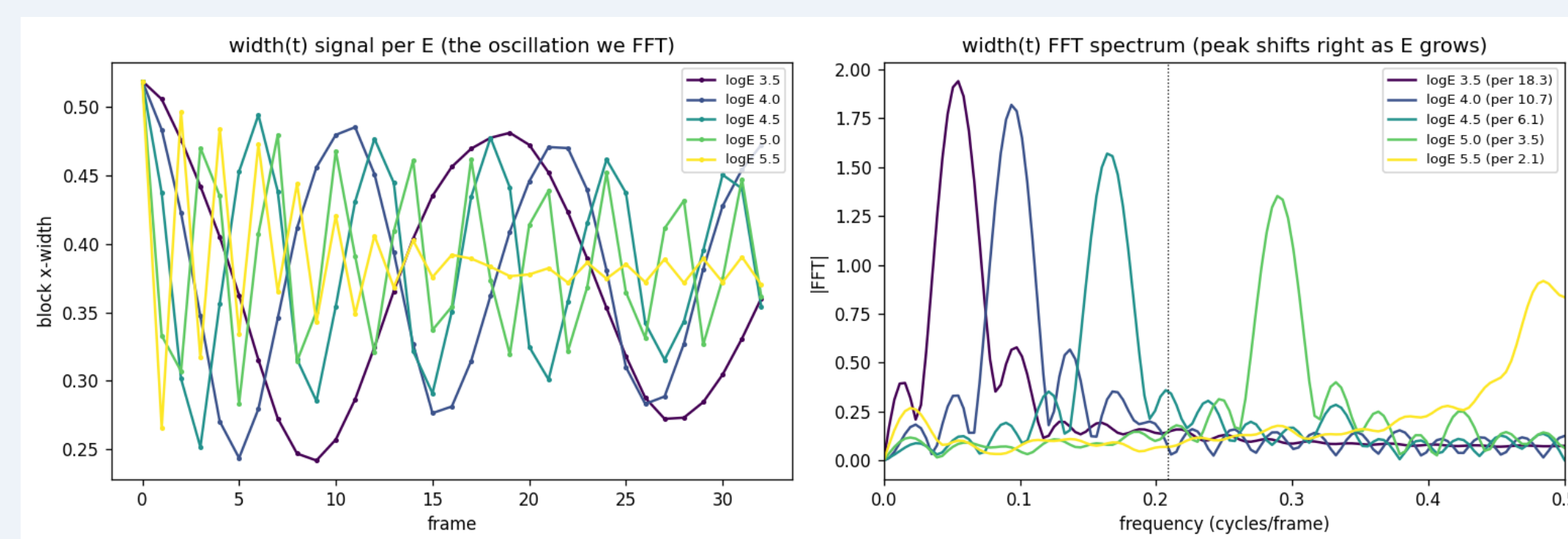
autograd/image-loss recovery in progress)

excitation	E channel	loss
displacement (release)	frequency	spectral $\propto \sqrt{E}$
force (drop/contact)	amplitude	time $\propto 1/E$

The excitation *type* decides which channel carries E , hence which loss can read it — **no universal loss**. This explains why gravity/contact benchmarks fit with a plain time/image loss, while pre-stress release needs a *spectral* loss (time-domain saturates into a beat plateau).

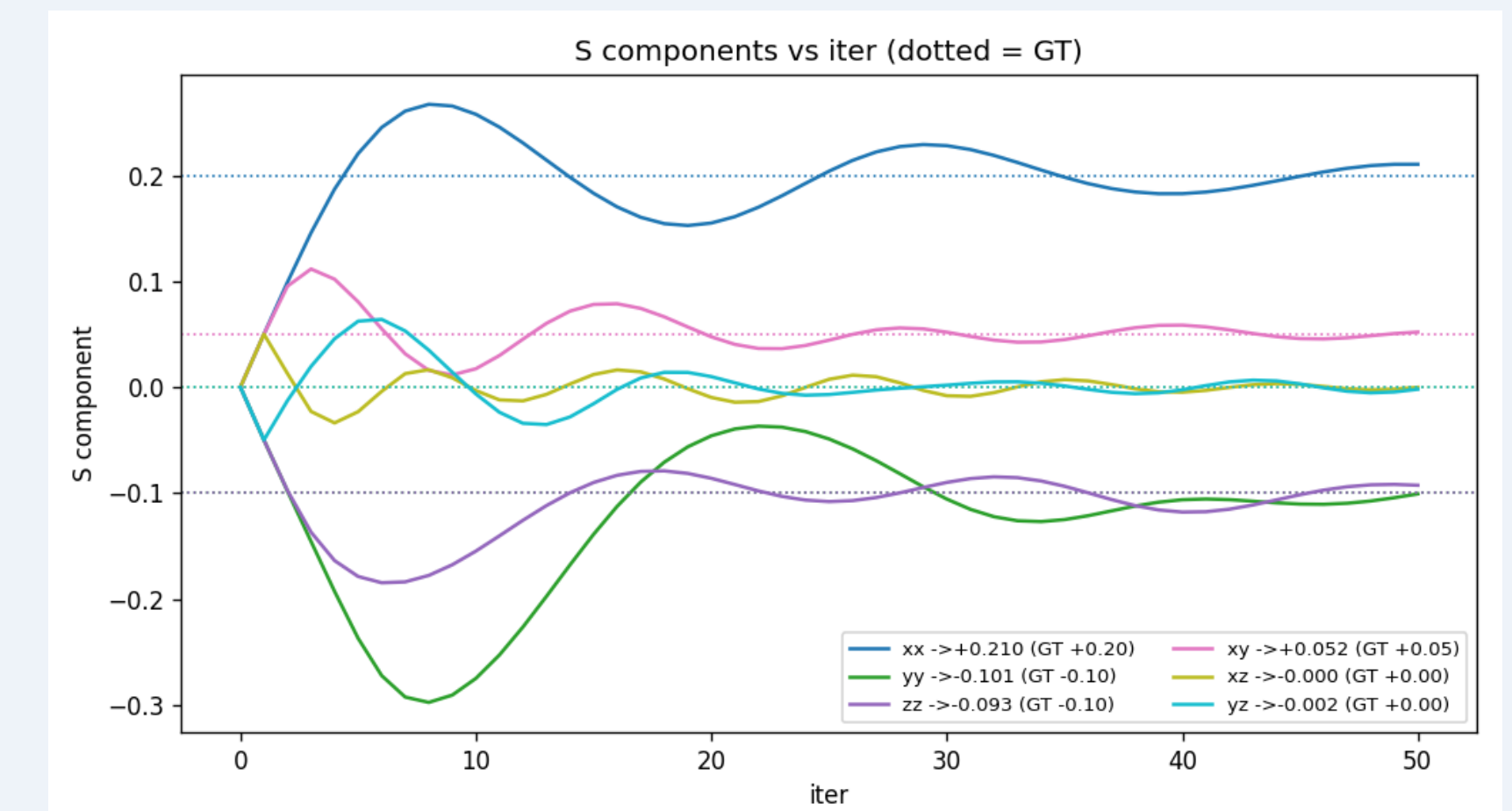


Complementary landscapes: time flat on *release*, spectral flat on *drop*.

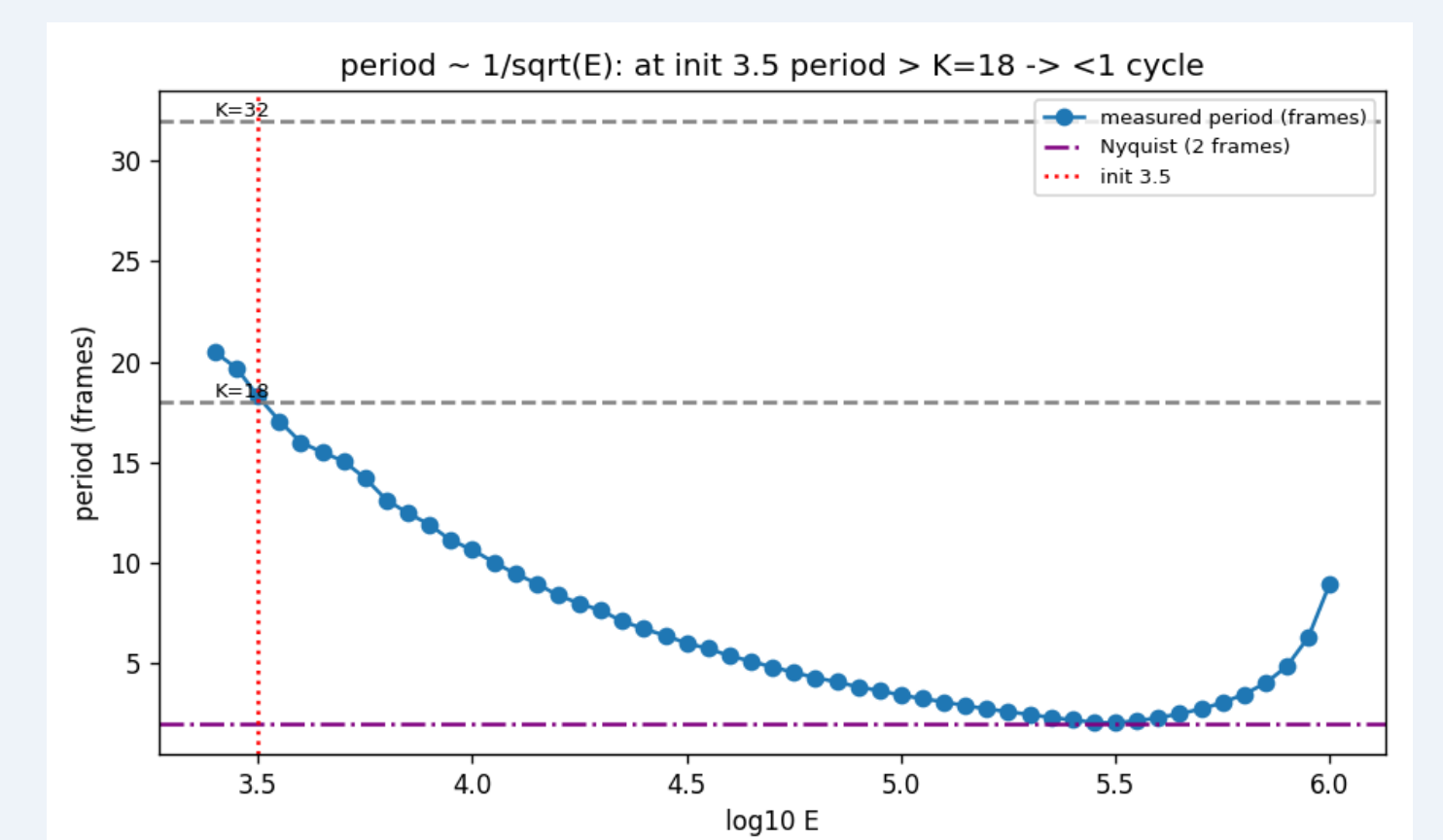


Displacement-excited: time-domain trajectories saturate; the spectrum stays discriminative.

F_0 is itself identifiable



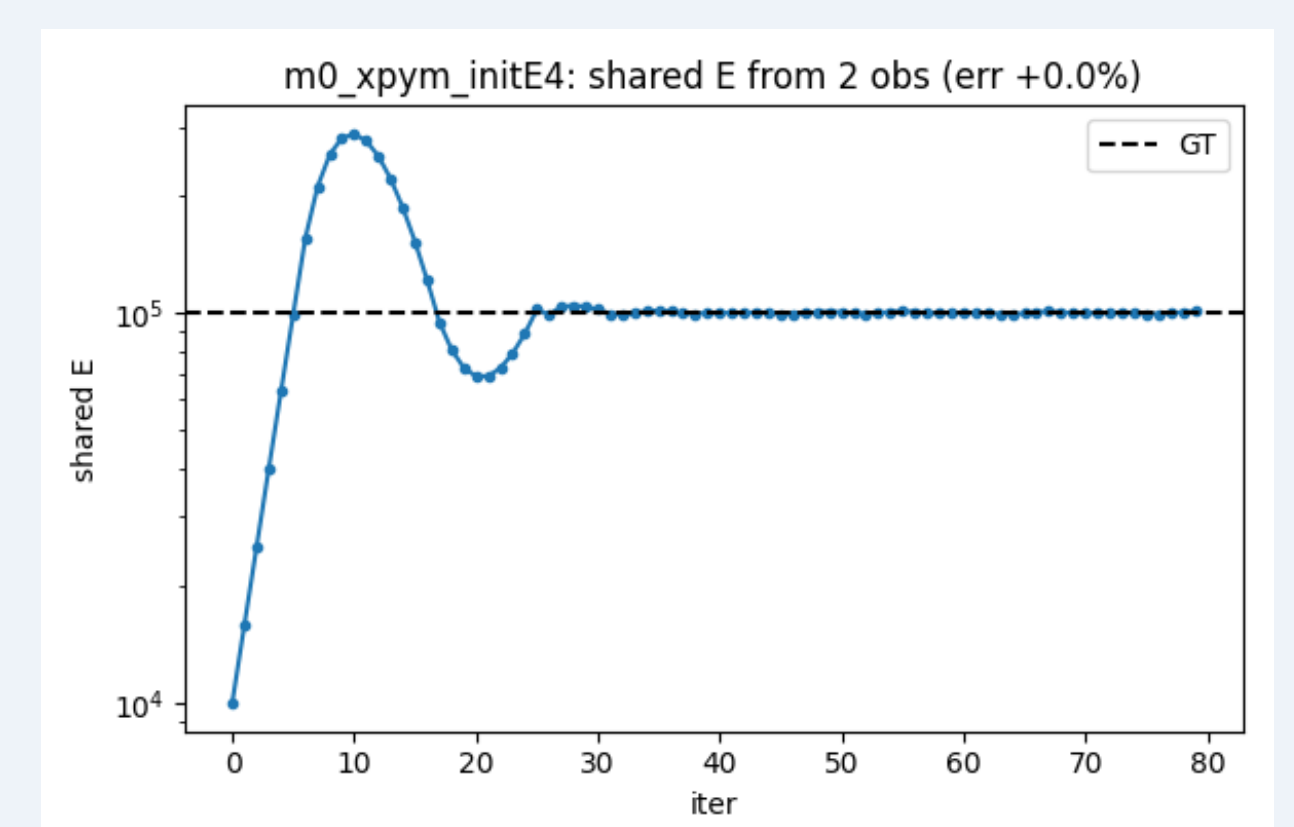
Global 6-DOF $V_0 = \exp(S)$: all components recovered, $\max|\text{err}|$ 0.010. Coarse 15-DOF field also recovered (one shallow bend degeneracy mapped).



Caveat: period $\propto 1/\sqrt{E}$ hits a Nyquist wall $\Rightarrow$ an aliasing false basin at high E .

Extensions & toward generative physics

- **Amortised multi-impulse:** one shared E from several excitations — **0.008%** (vs. 0.069% single). Shared material is learnable. (*oracle*)
- **Non-uniform E field** and F_0 -**conditioned** recovery (E on a pre-stressed rest state) — newest direction. (*oracle*) (forward+FD, synthetic)
- **Future:** amortise a *scene-conditioned physics prior* and sample plausible physics via VSD + reconstruction guidance.



Shared- E convergence across impulses.